Geometric Probability, Ballot, Birthday

colour key: navy = prose · teal = headings · yellow = key results · red = common traps

Continuous sample spaces

Uniformly chosen points in a region: probability is favourable area / total area, provided the joint distribution is uniform in the stated coordinates. Coordinate choice and sampling procedure matter.

WORKED EXAMPLE

Broken stick

Two independent uniform break points on a unit stick make three pieces. Triangle condition is that no piece is at least $1/2$. The region of valid break-point pairs has one quarter the total area, so probability $1/4$.

Bertrand ballot theorem

If $a > b$ votes are randomly ordered, the probability A is strictly ahead after every vote is $(a-b)/(a+b)$. Reflection of paths at their first diagonal contact counts the bad orderings. Strictly ahead excludes ties after the first vote.

Birthday collision

With d equally likely birthdays, independent and ignoring leap day, probability of no match among $n \le d$ people is $\prod_{j=0}^{n-1} (d-j)/d$; collision probability is one minus this. For $d = 365, n = 23$, about 0.5073.

COMMON TRAP

Model assumptions matter

Uniform birthdays, independence, and a 365-day year are approximations. The answer is conditional on this idealised model, not a statement about real populations.

Build state equations

For a process with finite memory, let p_s be the desired probability from state s. Condition on the next move: $p_s = \sum_t P(s \to t) p_t$, using boundary values for winning/losing states. For expectations, add one unit of cost per step: $E_s = 1 + \sum_t P(s \to t) E_t$.

WORKED EXAMPLE

Alternating first-head game

Fair coin; players alternate, first head wins. Let p be first player's win chance. She wins now with $1/2$ or both tosses fail with $1/4$ and state resets: $p = 1/2 + p/4$, hence $p = 2/3$.

Coupon collector

For n equally likely types, when j distinct types are collected the next-new-type chance is $(n-j)/n$. Summing geometric stage means gives

$$E[T] = n\left(1 + \tfrac{1}{2} + \cdots + \tfrac{1}{n}\right) = nH_n.$$

Memorylessness and runs

Geometric waiting time: $P(X > m+n \mid X > m) = P(X > n)$. There is no "due" effect. Expected tosses to obtain r consecutive heads obey a run-state recurrence; for a fair coin it is $2^{r+1} - 2$.

Gambler's ruin

Fortune starts at i units, absorbing barriers $0, N$, and each unit bet wins with probability p, loses with $q = 1-p$. Success probability solves $u_i = p u_{i+1} + q u_{i-1}$, $u_0 = 0, u_N = 1$:

$$u_i = \frac{1-(q/p)^i}{1-(q/p)^N} \quad (p \neq q), \qquad u_i = i/N \quad (p = q = 1/2).$$

INTUITION

Fair fortune is a martingale

Expected final fortune equals initial fortune i; final fortune is N with probability u_i and zero otherwise, so $Nu_i = i$.

Derangements by inclusion–exclusion

For n objects assigned uniformly to n positions, probability none fixed is

$$\sum_{k=0}^{n} \frac{(-1)^k}{k!},$$

so $D_n = n!\sum_{k=0}^{n}(-1)^k/k!$ and it tends to $1/e$. At $n=5$, $D_5 = 44$ and probability $11/30$.

WORKED EXAMPLE

Biased walk, exact value

Take $p = 3/5$, $q/p = 2/3$, barriers 0 and 10, start $i = 5$. Then $u_5 = [1-(2/3)^5]/[1-(2/3)^{10}] \approx 0.88364$. The edge makes success more likely than $1/2$; the boundary conditions still matter.

COMMON TRAP

Do not overgeneralize optional stopping

The elementary fair-ruin argument is safe for finite barriers and the process stopped at absorption. General martingale stopping claims require hypotheses; do not invoke an unrestricted optional-stopping theorem.

Derangement proof and a small check

Let A_i be the event that object i is fixed. Inclusion–exclusion gives $D_n = \sum_{j=0}^{n}(-1)^j\binom{n}{j}(n-j)! = n!\sum_{j=0}^{n}\frac{(-1)^j}{j!}$. For $n=3$, this is $6-9+2=2$; probability $1/3$. For $n=5$, $D_5 = 44$, probability $11/30$. The limit $1/e$ is approached with alternating partial-sum error bounded by the next reciprocal factorial.

Why averaging helps

For a random variable X, $E[X] = \sum_i E[I_i]$ when $X = \sum_i I_i$, regardless of dependence. This linearity powers coupon-collector stage sums and indicator proofs of expected collision counts. Variance does not share this unrestricted additivity.

What these extensions teach

The same ideas recur: complement for rare collisions, inclusion–exclusion for forbidden fixed points, state recursions for sequential games, and expectation decomposed into stages. These models add methods; they are not explicit JEE syllabus promises.